

Traditional flood detection and mapping methodologies have predominantly relied on segmentation-heavy pipelines and single-sensor inputs, typically using either optical or microwave data streams in isolation. While these approaches have demonstrated effectiveness in controlled or localized environments, they remain highly vulnerable to external environmental distortions. Factors such as wave interference, atmospheric scattering, surface reflectance variability, and reflection bias from mineral-rich surfaces (e.g., asphalt, urban infrastructure, and rocky substrates) significantly increase the susceptibility of these models to false positives. This limitation is further exacerbated in heterogeneous terrains where spectral ambiguity and backscatter variability degrade classification reliability. Existing studies acknowledge that regionally trained architectures can adapt to new geographical contexts when sufficient domain alignment is performed; however, such adaptations remain constrained by the quality, structure, and robustness of preprocessing pipelines.

Current flood extraction workflows are largely confined to rigid software ecosystems such as ENVI, SNAP, and IDL, where processing chains are tightly coupled and lack modularity. These monolithic implementations limit interoperability, scalability, and real-time deployment potential, particularly in distributed monitoring architectures and CubeSat-based observation systems. The absence of modular design restricts flexible upgrades, inhibits integration across heterogeneous sensor networks, and constrains deployment in cloud-based or edge-computing environments. Furthermore, Synthetic Aperture Radar (SAR)-based flood mapping introduces significant computational burdens due to multi-stage preprocessing requirements, including multi-looking, speckle filtering, radiometric calibration, and threshold-based segmentation. These computational demands restrict near real-time operation and prevent deployment in resource-constrained systems, which are increasingly relevant in disaster-response scenarios and autonomous satellite platforms.

Another critical limitation lies in the temporal dimension of flood monitoring. While existing studies effectively map flood extents, they provide limited support for continuous temporal change detection and automated multi-cycle pattern recognition across revisit periods of Sentinel-1 and Sentinel-2. Most current approaches emphasize static mapping rather than dynamic flood evolution modeling, reducing their effectiveness for early warning systems and real-time disaster response. Additionally, high-resolution satellite imagery often fails to capture flood peaks due to low revisit frequency, resulting in incomplete or misleading flood representations. This temporal sparsity introduces critical information gaps during peak disaster windows, weakening situational awareness and response planning.

Model generalization further remains a systemic challenge. Architectures trained on specific regions or sensors often fail to generalize across geographical domains and sensor modalities, limiting scalability and widespread applicability. Conventional convolutional neural network (CNN)-based models struggle with cross-domain transferability, particularly under varying atmospheric, surface, and hydrological conditions. Cloud cover further degrades optical imagery utility, making single-modality dependence fundamentally unreliable for continuous monitoring systems.

To address these limitations, recent research directions increasingly emphasize hybrid multi-sensor fusion architectures. Intelligent SAR–optical fusion frameworks offer resilience against cloud obstruction while improving surface discrimination reliability. Multi-sensor hybrid optical–microwave preprocessing pipelines enable robust flood detection by exploiting complementary physical properties of different sensing modalities. Integrating resource-efficient reinforcement learning (RL) models, complemented by statistical Bayesian inference frameworks, offers a promising pathway to reduce false positives while increasing detection precision under noisy environmental conditions.

Emerging solutions propose modular, lightweight preprocessing architectures where acquisition, filtering, calibration, resolution enhancement, and classification operate as independent yet interoperable modules. Such designs enable scalability, flexible upgrades, and seamless integration into distributed environmental monitoring systems, CubeSat constellations, and cloud-based platforms. Computational efficiency can be achieved through optimized noise-reduction algorithms, lightweight smoothing techniques, and threshold-based segmentation optimized for near real-time and edge-computing environments. Furthermore, machine learning–driven automated pattern recognition enables continuous temporal change detection and multi-cycle flood evolution tracking across Sentinel datasets.

Advanced AI-based data fusion approaches further propose integrating infrequent high-resolution satellite imagery with frequent coarse-resolution environmental data streams such as rainfall intensity, soil moisture, and hydrological indicators to generate high-temporal-resolution flood estimations. Vision Transformer–based architectures, trained on large-scale multi-regional datasets, offer improved generalization across geographic and sensor domains compared to traditional CNN models. These developments support the creation of automated, cloud-based real-time flood response systems capable of transforming heterogeneous satellite inputs into accessible, continuously updated flood intelligence products for disaster response agencies.

Collectively, these gaps highlight the need for a unified, modular, scalable, and computationally efficient multi-sensor preprocessing framework that integrates hybrid optical–microwave data fusion, AI-driven pattern recognition, and resource-efficient learning models. Such an architecture would enable real-time flood detection, multi-temporal change monitoring, reduced false-positive rates, and operational deployment across distributed satellite systems and cloud-based disaster response infrastructures.